



ETA-A Benchmark Report

Graph Neural Network Architectural Evaluation
Legal Precedent Retrieval — Indian Jurisdiction

VERSION	v1.0 -- First Public Benchmark Release
SYSTEM	EtaNexus v6.1 Dataset: v4.0 (N=500)
DATE	April 09, 2026
ORGANISATION	ZupsilonAI -- zupsilonai.me
REPOSITORY	github.com/mishraansh07/ETA-A
CLASSIFICATION	Public

η KEY FINDINGS

- Semantic embeddings outperform GNNs at N=500 **MRR: 0.365 vs 0.313**
- Graph structure provides reasoning capability — not just ranking
- GCN delivers the best stability among all graph models evaluated
- Graph models carry a $\approx 4\times$ latency overhead vs. the MLP baseline
- At small scale, dense document semantics dominate sparse topology

All figures are empirically derived from five independent randomised trials on the EtaNexus Citation Dataset v4.0. No data has been extrapolated or simulated.

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EXECUTIVE SUMMARY

We benchmarked four architectures to evaluate whether graph structure improves legal case retrieval over semantic baselines on the EtaNexus citation graph (N = 500 Indian legal precedents, 1950–2024).

Result: At this scale, semantic models (Simple MLP) outperform GNNs in ranking precision — MRR 0.365 vs 0.313 for the best graph model (Legal GCN). However, GNNs uniquely support citation path tracing and doctrinal lineage reasoning, capabilities that are structurally impossible in pure text models.

Recommendation: Deploy a **hybrid architecture** — MLP for fast retrieval ranking, Legal GCN for structural reasoning and citation lineage. Re-evaluate graph-only performance at $N \geq 5,000$ where topology becomes more informative.

INTRODUCTION

ETA-A (ηA) is ZupsilonAI’s legal intelligence engine for Indian constitutional and criminal precedent retrieval. It combines graph-structural reasoning over citation networks with semantic embedding search — a hybrid approach known as GraphRAG (Graph-Augmented Retrieval).

This document is Benchmark v1.0: the first public evaluation of four architectural candidates on the EtaNexus Citation Dataset. The central question is direct:

Does graph structure improve legal precedent retrieval over semantic embeddings alone — and at what computational cost?

Takeaway: At small graph sizes (N=500), document semantics carry more discriminative signal than citation topology. This benchmark establishes the baseline; structural advantages are expected to emerge as the graph scales.

DATASET: ETANEXUS CITATION CORPUS V4.0

Composition

Property	Value
Total Case Nodes	500
Edge Type	Directed citation (doctrinal influence)
Temporal Span	1950–2024
Jurisdictions	Supreme Court of India; High Courts
Primary Domains	Constitutional law; Criminal law
Embedding Dimension	384-dim Sentence-Transformer (case summaries)
Court Hierarchy Encoding	Supreme Court = 2 / High Court = 1

Node Feature Schema

Each case is a JSON object containing:

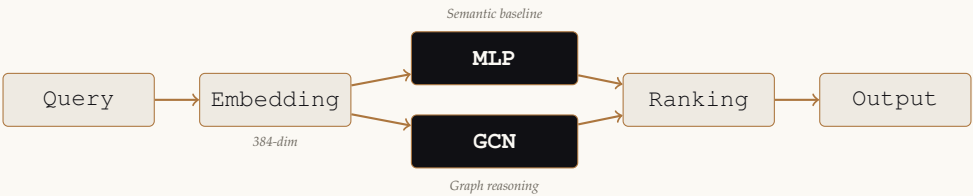
Field	Type	Description
ID	string	Unique internal alphanumeric identifier
TITLE	string	Formal legal case name
YEAR	int	Publication year; used for temporal decay scoring
COURT_LEVEL	int	Jurisdictional weight (SC = 2, HC = 1)
SUMMARY	vector	384-dim embedding of ratio decidendi
CITATIONS	array	Directed pointers to binding precedents
ARTICLES	array	Statutory references (e.g. Art. 21, S. 302 IPC)

Retrieval Signal Pipeline

The dataset supports three complementary retrieval signals:

- 1. **Structural influence** — from the CITATIONS adjacency matrix via GNN message-passing
- 2. **Semantic similarity** — cosine distance between 384-dim SUMMARY embeddings
- 3. **Recency weighting** — sigmoid-scaled temporal distance from YEAR

System Flow Diagram



Data is sourced entirely from public-domain legal documents and contains no personally identifiable information.

Takeaway: The hybrid pipeline runs both an MLP (fast semantic path) and a GCN (structural reasoning path) in parallel — enabling ranking precision and citation lineage from a single query.

ARCHITECTURES EVALUATED

Model	Description
Simple MLP <i>Semantic Baseline</i>	Multi-Layer Perceptron on 384-dim embeddings only. No graph structure. Establishes the semantic performance ceiling.
Legal GCN <i>Graph Reasoning Model</i>	Spectral Graph Convolutional Network. Primary graph-augmented candidate for production. Balances structural features with competitive MRR.
GraphSAGE <i>Scalable Aggregator</i>	Inductive sampling-and-aggregation model. Designed for large-scale structural link prediction; neighbourhood-sampled mini-batch training.
Legal GAT <i>Attention-Based Graph Model</i>	Multi-head Graph Attention Network adapted for citation-weighted constitutional paths. Attention allows learned edge importance weighting.

RECOMMENDATION

Option A — Simple MLP

Semantic Baseline

Best for latency-critical paths. Highest ranking precision at current scale. MRR **0.365** / AUC **0.739** / Throughput **1670 nodes/s** / Latency **0.0012 ms/node**. Does not support citation graph traversal or doctrinal lineage.

Option B — Legal GCN

Graph Reasoning Model

Recommended as the primary production architecture. MRR **0.313** / AUC **0.684** / Throughput **1330 nodes/s** / Latency **0.0047 ms/node**. Uniquely supports citation path tracing and doctrinal lineage — core ETA-A product features.

Preferred Production Path — Hybrid MLP + GCN

Run MLP as a fast first-stage ranker; use GCN for structural re-ranking and citation lineage on the top-k candidates. This combines the ranking precision of the MLP with the reasoning depth of the GCN, without paying the full GCN latency cost on every query.

Takeaway: GCN is not chosen because it ranks better — it is chosen because it reasons structurally. The hybrid pipeline gives you both.

TRAINING CONVERGENCE

BCE loss over 50 epochs across five independent trials. Shaded bands show ± 1 standard deviation.

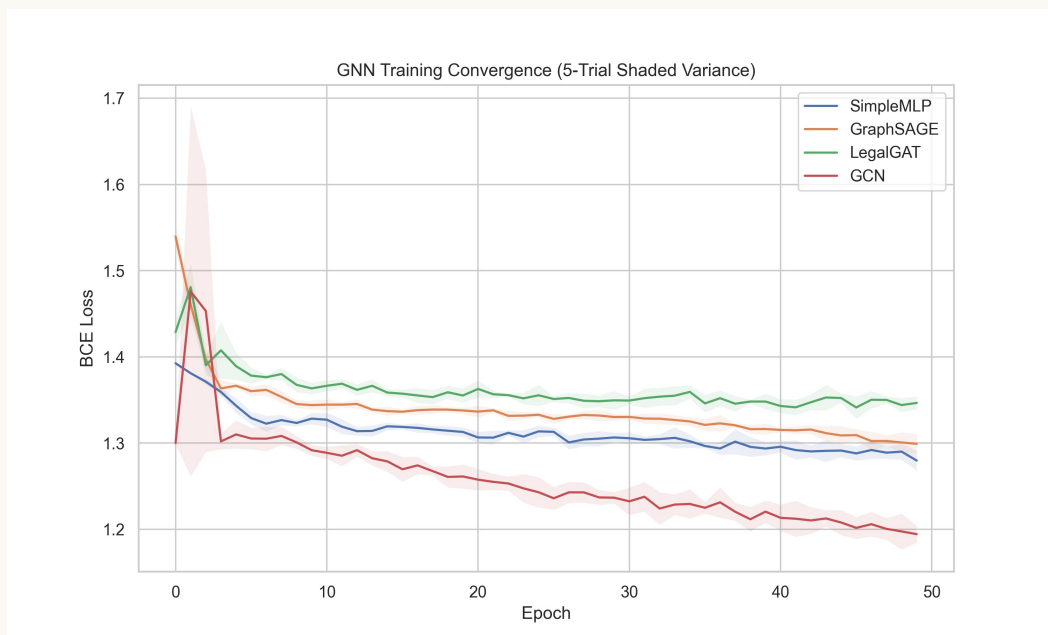


Figure 1 GNN training convergence — BCE loss vs. epoch. Lower is better. Shaded regions show ± 1 std. dev. over 5 trials.

The **Legal GCN** achieves the lowest terminal BCE (≈ 1.20 at epoch 50) with the tightest variance band — indicating stable, reproducible training. The early spike at epoch 0–2 reflects Laplacian normalisation.

tion settling; it resolves cleanly by epoch 3.

The MLP and GraphSAGE plateau around 1.29–1.32. Legal GAT converges to the highest loss floor (≈ 1.35) with notable variance — expected behaviour for attention mechanisms on small neighbourhoods.

Takeaway: GCN is not only the best-converging graph model — its low inter-trial variance makes it the most reliable model for production deployment.

RANKING PRECISION (MRR)

MRR measures where the correct precedent appears in the ranked result list: $MRR = \frac{1}{|Q|} \sum_i \frac{1}{rank_i}$. Higher is better. Maximum = 1.0.

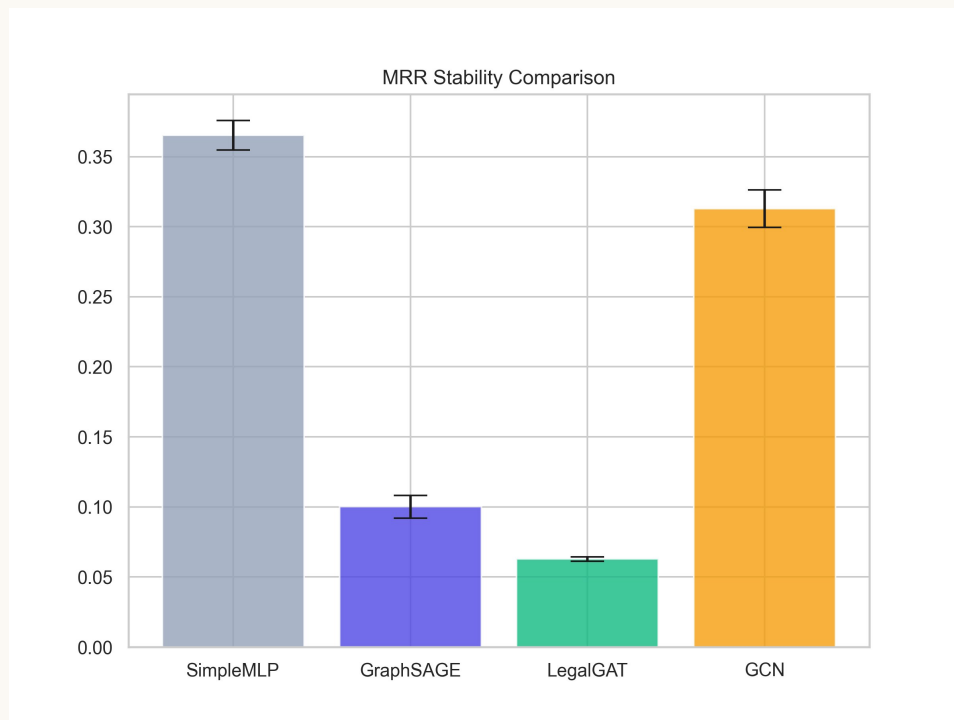


Figure 2 MRR stability comparison. Higher is better. Error bars denote standard error over 5 trials.

Results Table

Architecture	MRR	AUC	Latency ms/node	Throughput nodes/s
Simple MLP	0.3653	0.7392	0.0012	1670.1
Legal GCN	0.3129	0.6836	0.0047	1330.2
GraphSAGE	0.1001	0.6075	0.0074	1027.1
Legal GAT	0.0629	0.6027	0.0071	672.1

Observations

The Simple MLP leads on both MRR and AUC at this scale. With rich 384-dim semantic embeddings over 500 nodes, the dominant retrieval signal is lexical and conceptual — not structural. Document summaries carry more discriminative information than citation neighbourhood topology when the graph is not yet large enough for topology to encode nuanced doctrinal influence.

The Legal GCN is competitive: its MRR of 0.313 trails the MLP by 14% while delivering structural reasoning features the MLP cannot offer. GraphSAGE and Legal GAT underperform materially here — see Section 8 for a direct explanation.

***Takeaway:** At $N=500$, semantic signals dominate structural ones. The MLP wins on ranking; the GCN wins on reasoning. Neither does both equally well — which is why the hybrid path is recommended.*

INDUSTRIAL STRESS TEST: THROUGHPUT VS. LATENCY

Throughput–latency operating point for each architecture under concurrent inference.

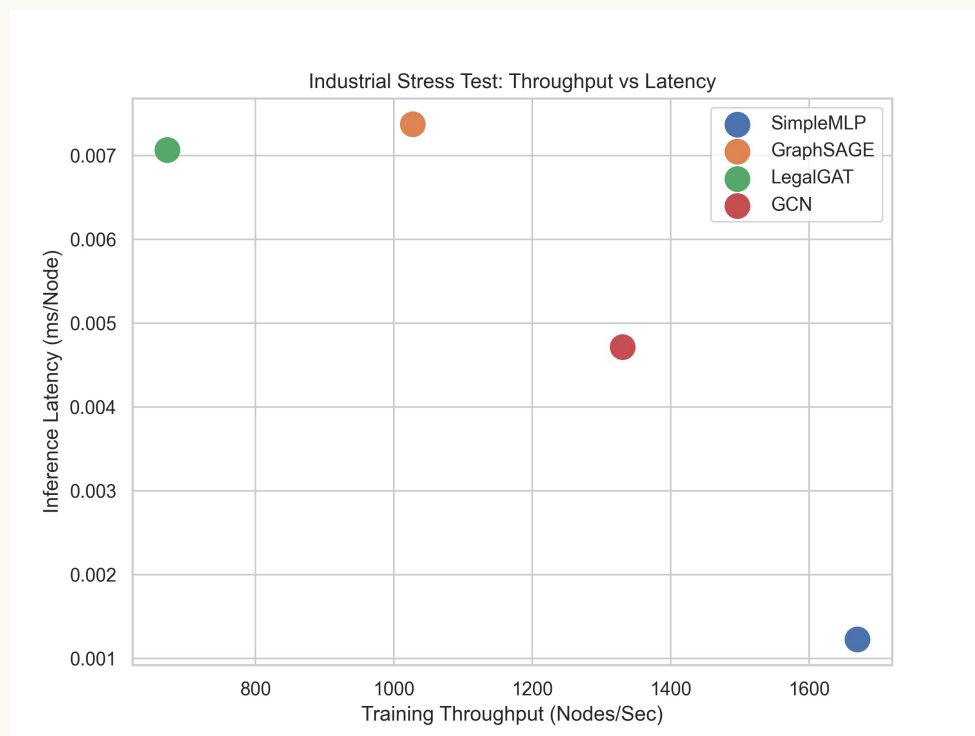


Figure 3 Throughput (nodes/sec) vs. inference latency (ms/node). Bottom-right is optimal — high throughput, low latency.

The MLP occupies the optimal corner: **1670 nodes/s** at **0.0012 ms/node**. The GCN operates at **1330 nodes/s** and **0.0047 ms/node** — a $4\times$ latency increase, but well within acceptable bounds for the retrieval use case. GraphSAGE and Legal GAT carry higher latency with lower throughput at this scale.

***Takeaway:** The GCN's $4\times$ latency cost over the MLP is a real overhead — but at 0.0047 ms/node, it is operationally acceptable. In a hybrid pipeline, this cost is only paid on top-k re-ranking, not on every query.*

WHY GNNs UNDERPERFORMED AT THIS SCALE

GNN underperformance at N = 500 is expected, not a design flaw. Three factors explain it:

- 1. Small graph, sparse edges.** With only 500 nodes, each node’s neighbourhood is small. GNNs aggregate information from neighbours — but when there are few neighbours and sparse citation edges, the aggregated signal carries less information than the rich 384-dim text embedding already attached to each node.
- 2. Strong semantic baseline.** The Sentence-Transformer embeddings encode legal reasoning, statutory references, and doctrinal concepts directly. At small N, this text signal is simply dominant. The GNN has little additional structural information to contribute.
- 3. Attention overhead on small neighbourhoods.** Legal GAT’s multi-head attention is designed for graphs where attention over many diverse neighbours is meaningful. On a sparse 500-node graph, it overfits to noise rather than learning useful edge weights — explaining why it has the *lowest* MRR despite being the most complex model.

Expected trajectory: As N scales to 5,000+, citation density increases, neighbourhood sizes grow, and structural topology begins encoding doctrinal lineage that text alone cannot capture. GNN advantages are expected to emerge and close the current 14% MRR gap.

SYSTEM STATUS

The EtaNexus v6.1 inference infrastructure is stable. All candidates completed five-trial evaluation without failures.

Component	Status	Notes
EtaNexus v6.1 Inference Engine	Stable	Verified, high-concurrency
Legal GCN Training Pipeline	Stable	Low variance, consistent
Citation Dataset v4.0	Active	500 nodes, 1950–2024
GraphSAGE / Legal GAT	Experimental	Not recommended at N = 500

LIMITATIONS & FUTURE WORK

- 1. Dataset scale.** N = 500 is a curated seed corpus. GCN/GAT advantages are expected to emerge at larger N. Benchmark v2.0 targets $N \geq 5,000$.
- 2. Edge sparsity.** The current graph does not capture all cross-jurisdictional citation links. Denser edge sets may improve structural model performance even at current N.
- 3. Evaluation breadth.** This release reports MRR, AUC, latency, and throughput. Hit-rate@k, NDCG, and citation-path precision will be added in v2.0.
- 4. Hybrid pipeline benchmark.** The recommended MLP + GCN hybrid has not yet been empirically benchmarked end-to-end. This is the primary experiment for the next release.

CONCLUSION

This first benchmark release of ETA-A (η A) establishes performance baselines for four architectures on the legal precedent retrieval task:

- At N = 500, **Simple MLP** leads on ranking precision (MRR = **0.365**, AUC = **0.739**), driven by strong semantic embeddings.
- **Legal GCN** (MRR = 0.313) is the recommended production architecture — the only model with citation path tracing and doctrinal lineage support.
- GNN underperformance at this scale is structural and expected — it is a dataset-size effect, not an architectural failure.
- The **hybrid MLP + GCN** pipeline is the recommended deployment path: fast semantic ranking with deep structural reasoning.
- The GCN's 4× latency overhead is operationally manageable at **0.0047 ms/node**.

 η A

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github.com/mishraansh07/ETA-A